HOW DO LLM's REASON ABOUT VOTER BEHAVIOR?

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LLMs are everywhere, but can we trust them with something as important as an election?

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Method, Analysis, Result

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Summarize everything

INTRODUCTION

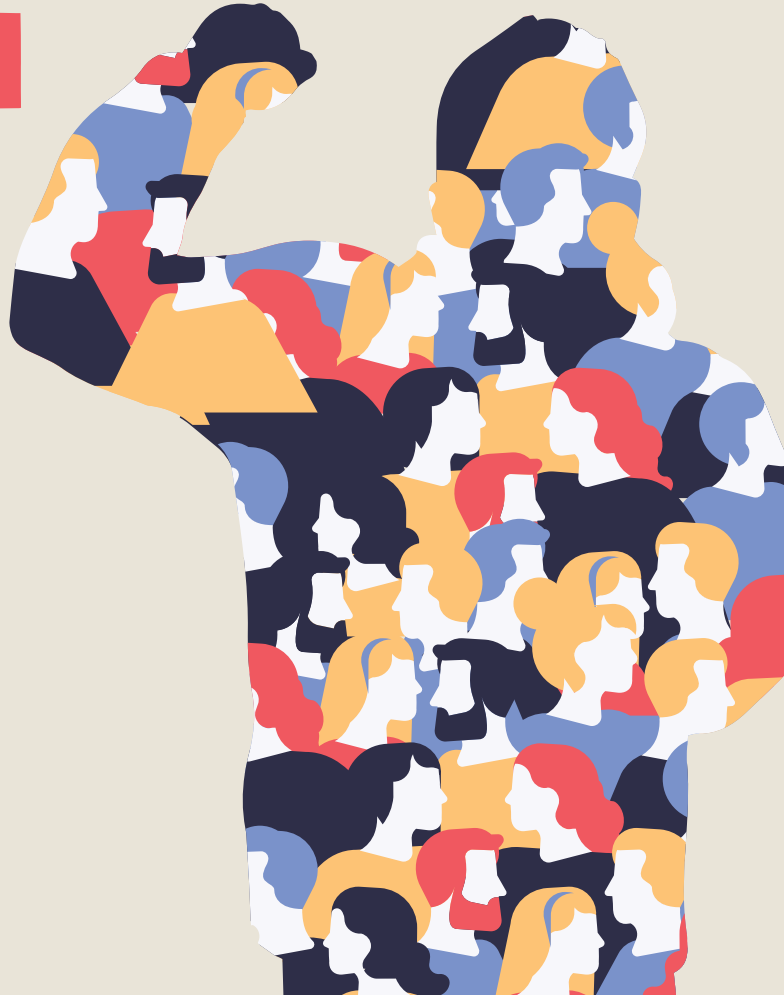
- Large Language Models (LLMs) are increasingly used to answer questions about politics and elections
  (An Empirical Study of Collective Behaviors and Social Dynamics in Large Language Model Agents, 2026)

- LLMs may infer voter behavior from demographic information (age, gender, location)

- However, these predictions may rely on stereotypes or simplified patterns

- It is important to evaluate whether these predictions are **accurate and consistent**

- **Research goal:** analyze how an LLM reasons about voter behavior using simulated voter data



EXPERIMENTAL DESIGN

SIMULATE REAL VOTERS

1. Sample a WA county
    a. Weight: P(county)
2. Sample age group and gender
    a. Age weight: P(age group|county)
    b. Gender weight: P(gender|county)

(Reports, Data, and Statistics — WA Secretary of State, 2024)

SIMULATE FURTHER COVARIATES

Covariate Groups:

- Demographics
    - race, age, gender, urbanicity
- Family
    - marital status, number of kids, religion
- socioeconomic status
    - education, income

Uniformly random sample from covariate levels

EXPERIMENTAL DESIGN

QUESTIONS

1. General voting behavior
2. Specific candidates (2024 General Election Voters' Guide – WA Secretary of State, 2024)
   a. Federal
   b. Washington State
3. Specific Measures (2024 General Election Voters' Guide – WA Secretary of State, 2024)

Consistency only

1. Is behavior easy to predict
2. Political behavior (Pew Research Center, 2021)

EXPERIMENTAL DESIGN

MODEL & INPUT

Model : Google Gemini *gemma-3-27b-it*

Input : Voter data (*column order*): *data*. For each voter in the voter data, *question*? Respond with only *response options*. *Rules*

Example question and response options:

Voter data (race, voter ID, urbanicity, age, gender): Prefer not to say, 74, urban, 18-24, female; Multiracial, 14, rural, 18-24, nonbinary. For each voter in the voter data, how would the voter vote on the following: Initiative Measure No. 2117 *measure details here*. Should this measure be enacted into law? Respond with one word, yes no.

Output : 74. yes \n 14. no

RULES

Accuracy :

Can only return unknown sparingly, do not omit any voter, produce EXACTLY 100 voters with voter id's 1 to 100, output MUST be sorted by voter id in order given, do not fabricate voter id values only use those provide, output format: voter id. Prediction

Consistency :

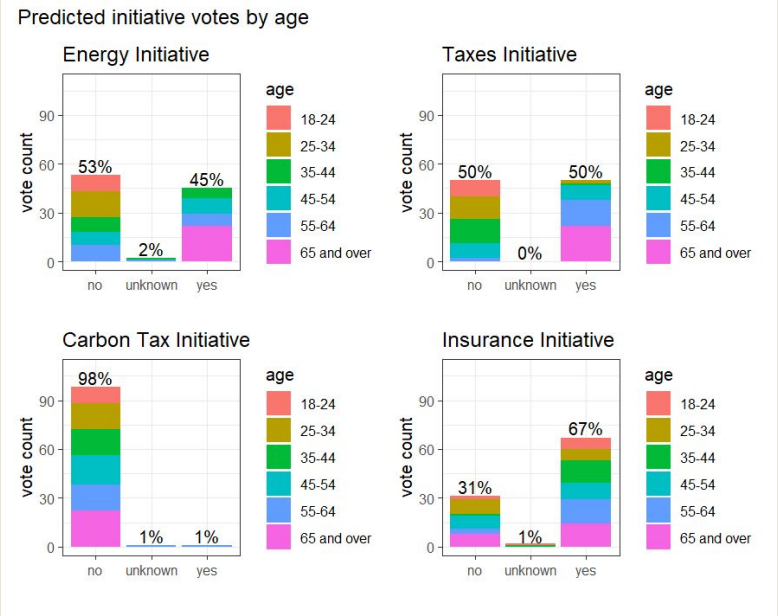
Rules (MUST follow all): Can only return unknown sparingly, do not omit any voter, produce EXACTLY 100 predictions, return only the 100 prediction lines with no intro or explanation, for line k, use voter ID from row k in the voter data above, do NOT sort, group, or reorder by voter ID; keep the exact row sequence as provided, do not fabricate voter ID values, Format each line exactly: voter ID. prediction.

ACCURACY EDA

vote_democrat	vote_republican		
	no	unknown	yes
no	6	0	44
unknown	0	1	0
yes	26	0	23

what_party	vote_democrat		
	no	unknown	yes
Democrat	20	1	26
Republican	30	0	23

what_party	vote_republican		
	no	unknown	yes
Democrat	21	1	25
Republican	11	0	42



ACCURACY ANALYSIS

Goal: Measure **how accurate** the models responses are when provided simulated Washington voter data

Method:

1. **Calculate Expected Probabilities**
   a. P(age, gender|county) = P(age|county) · P(gender|county)
   b. E[prediction, county, age, gender] =
      P(age, gender|county) · count(prediction|county) · 100/N
2. **Conduct exact multinomial goodness-of-fit test**
   a. On entire data set
   b. On aggregated data sets
3. **Calculate Kullback-Leibler divergence**
   a. On entire data set
   b. On aggregated data sets

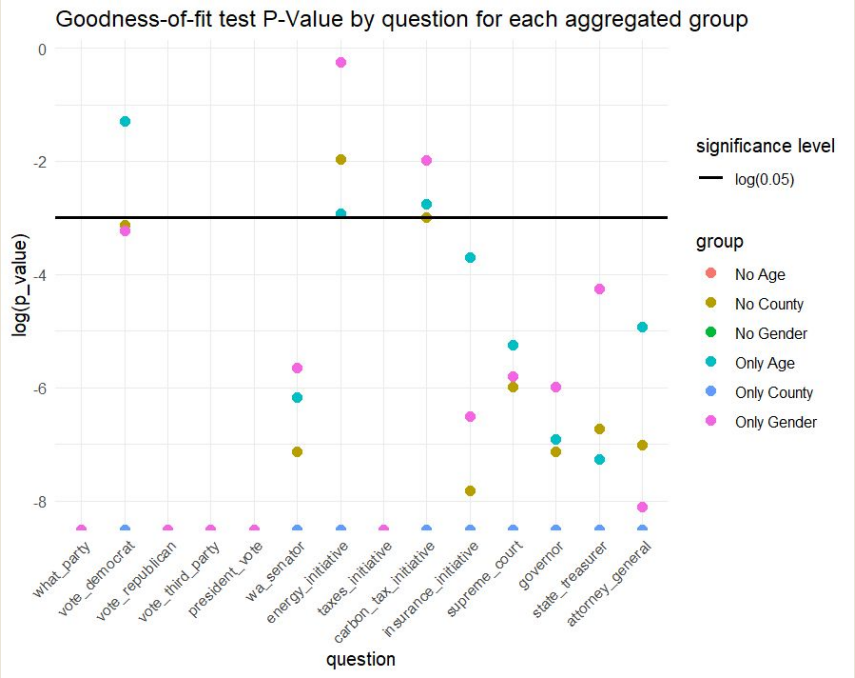
ACCURACY RESULTS

GOODNESS-OF-FIT

Entire data set: all p-values << 0

Aggregated data sets insignificant p-values:

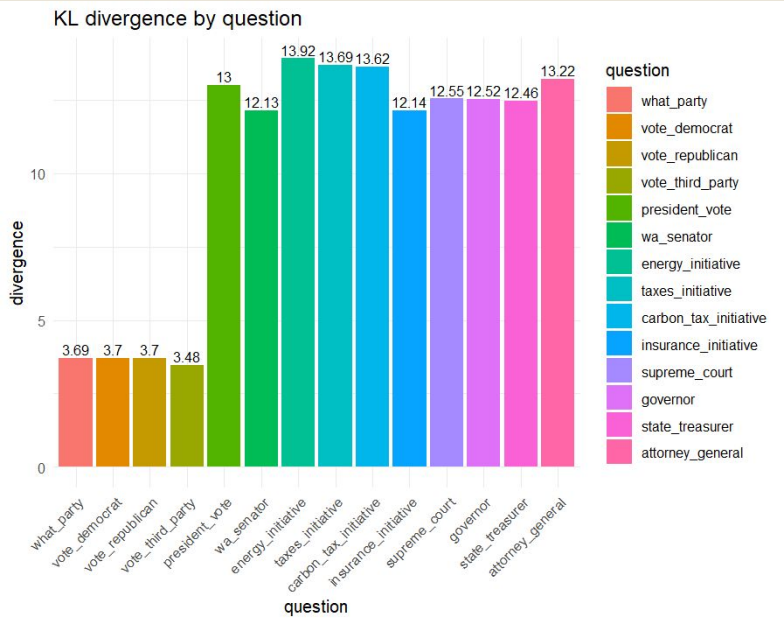
- Only age
  - Vote_democrat; energy_intitaitve; carbon_tax_initiative
- Only gender
  - energy_initiative; carbon_tax_initaitive
- No county
  - energy_initiative

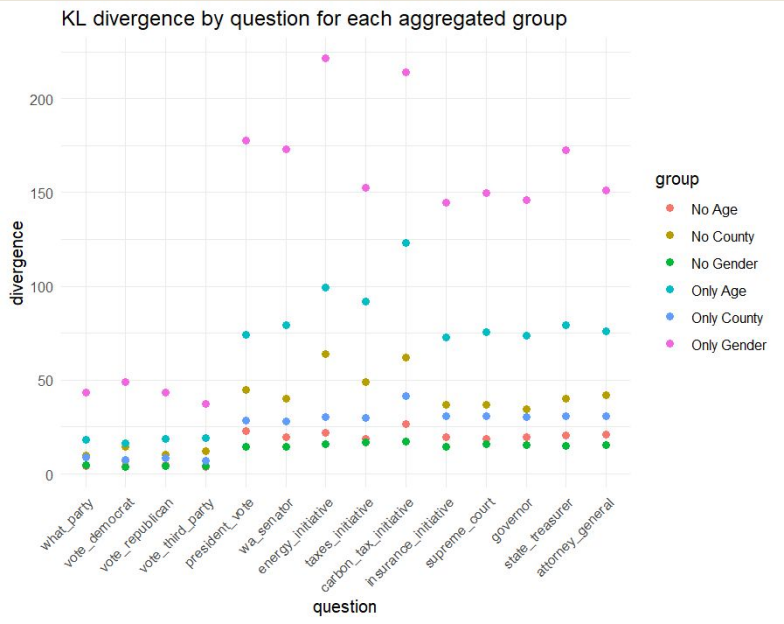


ACCURACY RESULTS

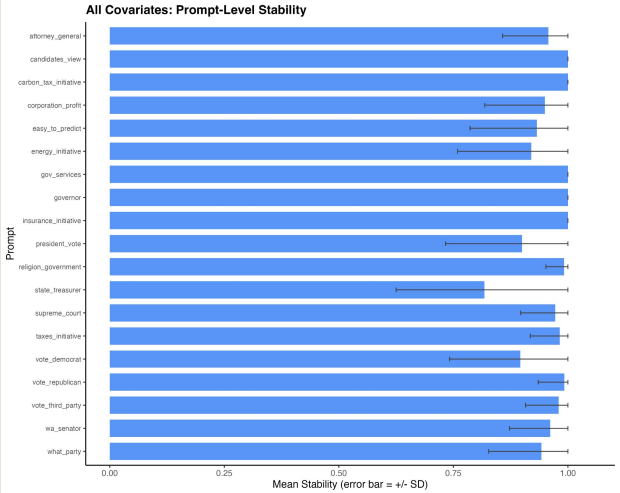
KL DIVERGENCE

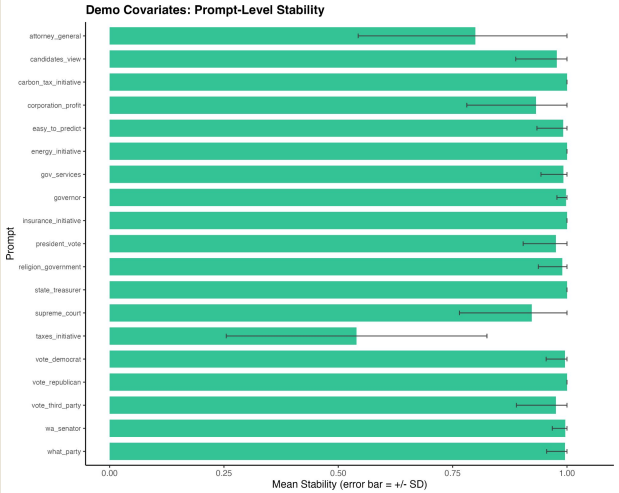
Lower divergence for general questions; higher divergence for specific questions

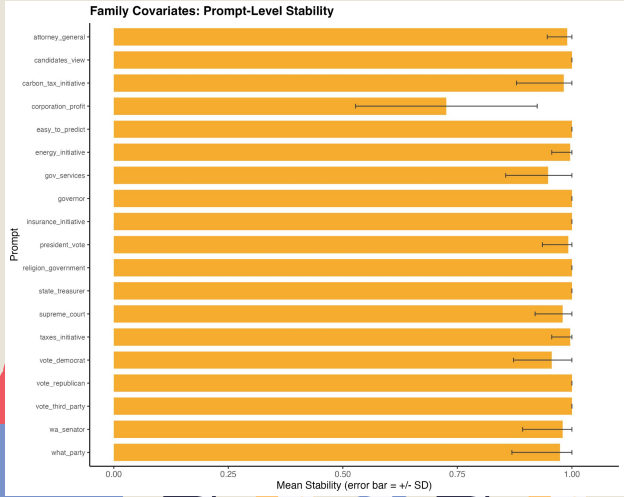


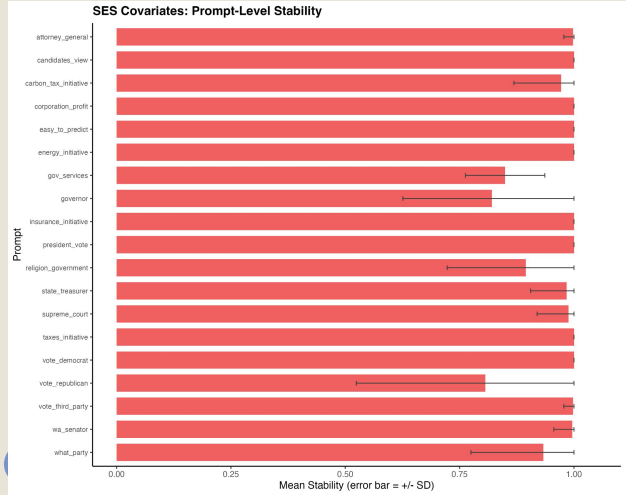


CONSISTENCY EDA









All Covariates

- Mean stability: **0.9576**
- **Lowest variability** (SD = 0.0481)
- Most **uniform across prompts**

Demo Covariates

- Mean stability: **0.9518**
- **Highest variability** (SD = 0.1105)
- More **sensitive to prompt wording**

Family Covariates

- **Highest mean stability**: **0.9747**
- Moderate variability (SD = 0.0622)
- Most **stable on average**

SES Covariates

- Mean stability: **0.9600**
- Moderate spread (SD = 0.0661)
- Strong but slightly less uniform than full model

CONSISTENCY ANALYSIS

Goal: Measure **how stable** the model's responses are when the same prompt is repeated multiple times for the same voter profile.

Method:

1. **Sample Voters:** 100 voters for each covariate group -> 4 df
2. **Prompting:** 19 prompts per voter -> Each prompt runs **5 times.**
3. **Response Encoding** : depends on number of answer options

   (e.g., Pete=1, Nick=0 or Yes=1, No=0, Unknown=2)
4. **Response Vecto** r: 5 response per voter prompt (e.g, [1,1,1,1,0])
5. Classify **Stability Categories**
6. Compute **Stability Score** and **Consistency Levels**
7. **Summary Statistics** -> overall mean, standard deviation (across prompts)
8. Fit **Ordinal Regression** Model

CONSISTENCY EXAMPLE

- **Prompt 1 Answer Options** : Pete or Nick -> **Pete = 1, Nick = 0**
- **Response of Voter 1** : Pete, Pete, Pete, Pete, Nick -> **[1,1,1,1,0]**
- **Stability Classification** :
  - Perfectly Stable (5-0)
  - **Slightly Unstable** (4–1)
  - Unstable (3–2)
- **Stability Score:**

$$\text{stability_score}_i = \frac{\max\{n_{\text{pete},i},\ n_{\text{nick},i}\}}{5}.$$

- Consistency Levels:
  - 1.0: High
  - **0.8: Medium**
  - 0.6: Low

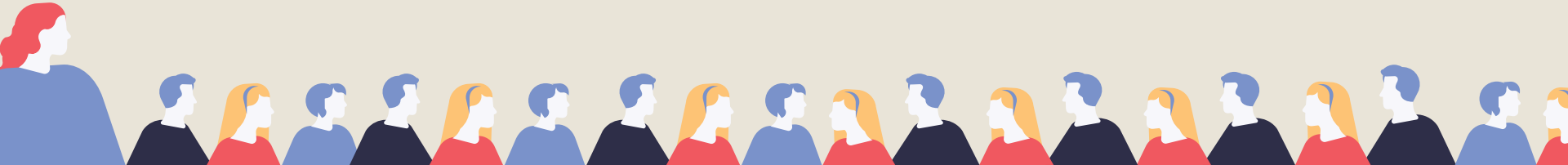
CONSISTENCY RESULTS

Summary

Covariate Group	Prompts	Overall Mean	Overall SD
all_covariates	19	0.9576	0.0481
demo_covariates	19	0.9518	0.1105
family_covariates	19	0.9747	0.0622
ses_covariates	19	0.9600	0.0661

Result Interpretation:

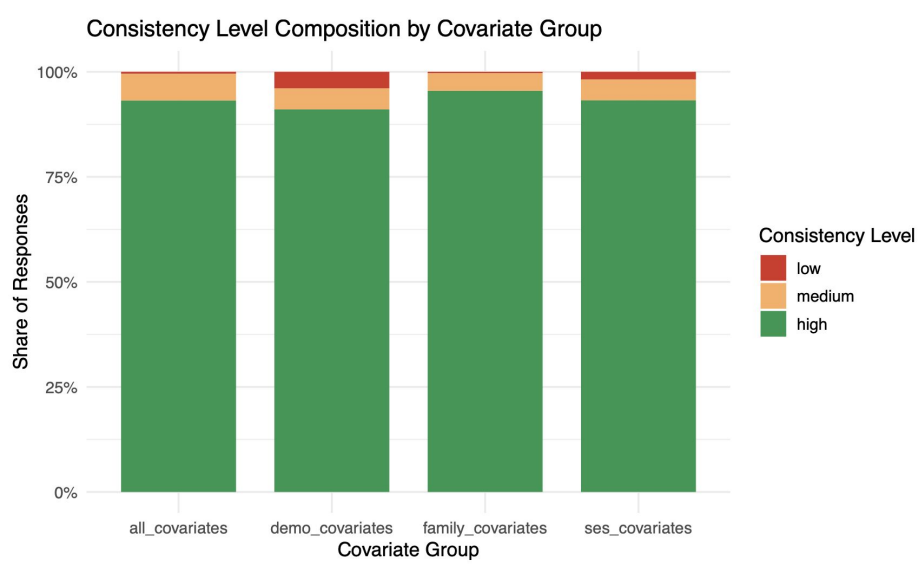
- all_covariates -> highest mean, lowest SD
- More complete covariates -> more stable & less random responses



CONSISTENCY RESULTS

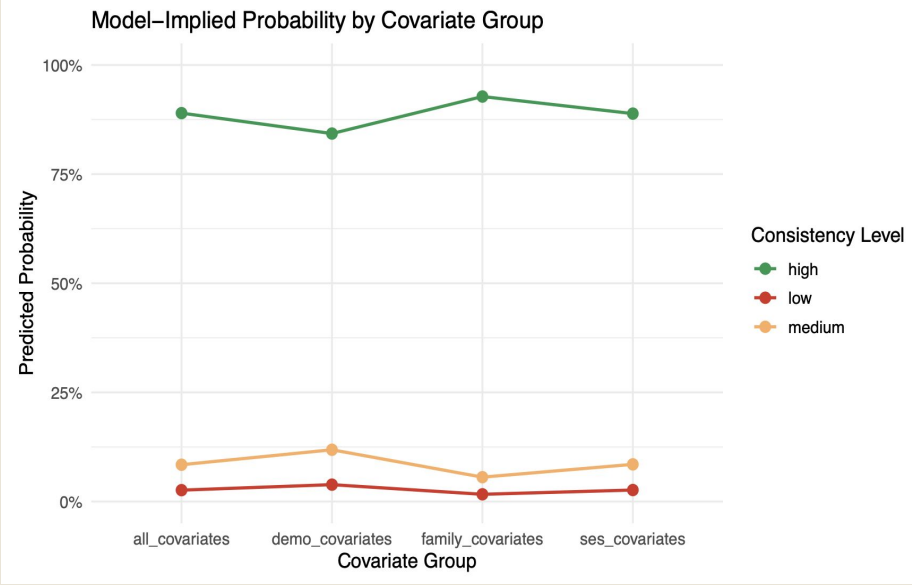
Ordinal Regression:

consistency_level ~ all_covariates + prompt



P (consistency_level | covariate_group):

"How likely is the LLM's response to be stable for each covariate group?"



CONCLUSION

CONSISTENCY

- **Family covariates performed best:** highest odds & highest predicted probability.
- **Covariates matter:** *family covariates* show the highest stability, while *demographic-only* prompts are more variable and sensitive to wording.
- **Implications:** stability suggests systematic reasoning, but: ***consistency ≠ accuracy*** **or lack of bias** , highlighting the need to evaluate both reliability and potential bias in LLM political predictions.

ACCURACY

- Not able to predict voter behavior with a high degree of accuracy
- Generally more reliable for general questions, less for precise questions
- Over-reliance on specific covariates such as age

Results also indicate LLM seems to understand there is a relationship between covariates and voting behavior.

THANK YOU!

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